the Journal of Strength and Conditioning Research™

The Relationship Between Relative Rest and Team Performance Across Competitive Seasons in the National Football League

Anthony G. Pinzone, Yousuf Abu-Amara, Ryan W. Gant, Jennifer Rivera, Emily C. Tagesen, Jacob E. Barkley, and Adam R. Jajtner

Exercise Science and Exercise Physiology, Kent State University, Kent, Ohio

Abstract

Pinzone, AG, Abu-Amara, Y, Gant, RW, Rivera, J, Tagesen, EC, Barkley, JE, and Jajtner, AR. The relationship between relative rest and team performance across competitive seasons in the national football league. *J Strength Cond Res* XX(X): 000–000, 2025—This investigation aimed to assess the relationship between relative rest index (RRI) and team performance across multiple seasons of National Football League (NFL) competition. To examine this relationship, all NFL regular season games from the 1970–71 to 2022–23 NFL seasons (N = 12,290) were analyzed with rest intervals calculated for each team on a game-to-game basis as the number of days between current and previous games. Next, RRI was calculated as the discrepancy in rest intervals between competing teams for all games and averaged for all teams across all seasons with teams categorized by season average RRI (<-0.70; -0.7 to -0.5; -0.5 to -0.25; -0.25 to 0; 0 to 0.25; 0.25 to 0.5; 0.5 to 0.7; >0.70). Win percentage and playoff depth (lost in wild card round; lost in divisional round; lost in conference championship; lost in Super Bowl; won Super Bowl; did not make playoffs) quantified team performance. Pearson product correlations were implemented to assess the relationship between season average RRI and win percentage. One-way analyses of variance assessed differences in win percentage across RRI categories and RRI across playoff depth categories. A trivial negative correlation between RRI and win percentage across all seasons was noted. Moreover, win percentage did not differ across RRI categories, while RRI was attenuated among teams that lost in the Super Bowl and divisional round compared with teams that did not qualify for the playoffs, suggesting that RRI has no significant effect on team performance with competitive eras of NFL competition collated.

Key Words: American football, rest, performance

Introduction

Since its establishment in 1920, the National Football League (NFL) has gained immense popularity across the United States and around the world, averaging 16.7 million viewers per regular season game in 2022 (2). Currently, the NFL season is structured in an 18-week format, with each team playing a total of 17 games with 1 bye week that occurs between weeks 5 and 14 (14). This differs from previous years such as 1935–1978, where teams only played between 10 and 14 games per season, and 1978–2020, where teams played 16 games per season (1). Accordingly, NFL athletes participate in a physically demanding contact sport during the course of a 17-game season, independent of postseason competition.

Multiple investigations have demonstrated alterations in markers of muscle damage, inflammation, and performance throughout a season of competition across various sports (7,8,16). A single week of soccer competition significantly increases circulating creatine kinase and granulocyte counts, impairs sprint and jump performance, and reduces isokinetic knee flexion and extension peak torque up to 6 days postcompetition (16). Similar patterns have been observed among collegiate American football players with acute increases in markers of muscle damage and attenuations in athletic performance demonstrated both during and after a single game (8), and chronically throughout an entire season (7). This suggests that elite football athletes accumulate stress during the course of a season, indicating that greater rest between games may be advantageous. In addition, athletes perceive rest as vital for maintaining performance throughout a competitive season, because reductions in rest with concomitant increases in workload result in decrements in perceived readiness to perform (13). In support of this notion, elite soccer athletes reported reduced scores on the Recovery-Stress Questionnaire for Athletes and increased incidence of burnout and depressive symptoms as the season progressed, despite no attenuation in maximal running velocity or vertical jump performance (5,18). A review by Hrysomallis (10) suggested that upper body strength and power vary heavily throughout a competitive football season, but may be better maintained with a high-intensity, low-volume resistance training configuration that maximizes rest between training sessions. Moreover, multiple studies (15,17) have demonstrated that acute psychological stress can attenuate exercise and sports performance across multiple athletic populations. As such, it remains clear that athletes perceive rest as important, and that reduced rest between games during a competitive season may further attenuate athletic performance and athlete well-being.

Owing to the scheduling format of NFL games from week to week, athletes can compete at any time of day on a Sunday, and in the evening on Thursday or Monday, and on Saturday throughout the day during later weeks of the season (14). The scheduling

Address correspondence to Adam R. Jajtner, ajajtner@kent.edu.

Journal of Strength and Conditioning Research 00(00)/1–5

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configuration of NFL games and the travel requirements imposed on athletes often result in a discrepancy in rest time from the previous game between 2 competing teams for a given week. This discrepancy, which has been referred to as relative rest index (RRI), is the difference in rest time from the preceding game between 2 competing teams. There seems to be substantial interteam variability in RRI as prior data from our laboratory indicated the average season relative rest ranged from −1.13 to 1.34 days during the 2022 season, despite both teams being in the same division (12). Furthermore, a positive correlation ($r = 0.339$) was observed between RRI and Win/Loss Ratio percentage, which when adjusted to an R^2 value indicated that 11.4% of the variability in Win/Loss Ratio was explained by RRI (12).

Presently, RRI is topical in the NFL, because the NFL Players Association pushed back against the idea of "flexing" or shifting scheduling in prime-time games on short notice; specifically, allowing Thursday night games to be eligible for "flexing" during a given week (4). The schedule of the NFL season imposes a large physical demand from the number of games played alone, although prior data indicate that the rest duration also likely affects performance (12). Although the correlation between RRI and Win/Loss Ratio has been evaluated during the 2022 NFL season (12), no prior work has assessed this relationship during previous NFL seasons. Therefore, the purpose of this investigation was to evaluate the relationship between RRI and Win/Loss Ratio during all previous NFL seasons, within the Super Bowl era (dating back to 1970). We hypothesized that RRI and Win/Loss Ratio would be positively correlated throughout all seasons. We also hypothesized that teams with greater average RRI would display greater Win/Loss Ratio across all seasons. Finally, we hypothesized that teams who progressed furthest in playoff competition would display greater mean RRI values across all seasons.

Methods

Experimental Approach to the Problem

To examine the influence of RRI on Win/Loss Ratio, we assessed the relationship between RRI and Win/Loss Ratio for all teams during the 1970–71 to 2022–23 NFL seasons. Moreover, RRI values were also compared across varying categories of playoff performance from not qualifying for the playoffs to winning the Super Bowl during a given season. In addition, to understand the effect of varying RRI on Win/Loss Ratio, 8 RRI categories were created based on specific RRI ranges in descending order with RRI for all teams for each season categorized accordingly.

Subjects

To evaluate the effect of RRI on Win/Loss Ratio, a total of 12,290 NFL regular season games played between the start of the 1970–71 and end of the 2022–23 seasons were analyzed. Subsequently, each individual NFL franchise for each season was treated as its own team for a total of 1,569 teams. Retroactive analysis was terminated at the 1970 season because no record of specific game times before 1970 is publicly available from profootballreference.com or NFL.com. Approval for this investigation was sought from the Kent State University's IRB, however, the IRB determined that this investigation did not require IRB approval.

Procedures

Data Acquisition. All data were directly imported into Microsoft Excel (Microsoft Corporation, Redmond, WA) from pro-football-reference.com, a website that maintains an up-to-date database of NFL statistics and record of every game played. Data were imported on a season-by-season basis with the game week, day of the week, date, time, winner, loser, and box-score provided from the website for each game. A total of 12,290 of regular season games ranging from September 18, 1970 (week 1 of 1970–71 Season) to January 8, 2023, were used for analysis. The final list of games was manually spot checked against the existing database of games that is available on NFL.com. In the event that dates and times of postponed or rescheduled games were not accurately reflected on profootballreference.com, correct dates were verified on NFL.com and manually added.

Determination of Relative Rest and Data Analysis. Relative rest index was calculated as the discrepancy in amount of time between prior and current games between competing teams and can be expressed as

$$\begin{aligned} \text{RR1(Days)} = {} & (\text{Time of Previous Game}) \\ & - (\text{Time of Opponent Current Game} \\ & - \text{Time of Opponent Previous Game}). \end{aligned}$$

For example, if 2 teams playing each other at 4:25 PM EST on a Sunday had their prior games on 1:00 PM EST and 4:25 PM EST the previous Sunday RRI would be expressed as 0.146, or −0.146, respectively. Notably, all times between current and prior games were strictly expressed in EST. Time zone discrepancies in game location were not factored in when calculating RRI for each game. Although time between final preseason and week 1 regular season games differed from team to team, all teams were assigned the same time between prior and current game (0) during week 1 because demands and player rotations vary heavily between preseason and regular season games. As such, all teams had no difference in RRI during this week. Average RRI was then generated for each team individually for each season by averaging RRI from across all weeks of regular season play. Win/Loss Ratio was also calculated for each team for all seasons as the number of wins per season divided by the total number of games per season.

To examine team success within seasons, teams were placed into playoff performance categories: (a) lost in wild card round; (2) lost in divisional round; (3) lost in conference championship; (4) lost in Super Bowl; (5) won Super Bowl; (6) did not make playoffs. Finally, each NFL team was placed into 8 categories

Table 1

Total number of occurrences of teams in each RRI category across all seasons and across each separate periods of NFL competition.*†

	Total	2010–2023	2000s	1990s	1980s	1970s
>0.70	30	16	6	6	0	2
0.5 to 0.7	94	29	32	31	0	2
0.25 to 0.5	229	72	55	55	27	20
0.00 to 0.25	481	98	82	70	115	116
0 to −0.25	403	86	59	47	108	103
−0.25 to −0.50	206	72	44	45	25	20
−0.50 to −0.70	57	21	14	16	3	3
< −0.70	69	22	26	19	1	1

*RRI = relative rest index.

†Total number of occurrences of teams in each RRI category across all seasons and across each separate periods of NFL competition.

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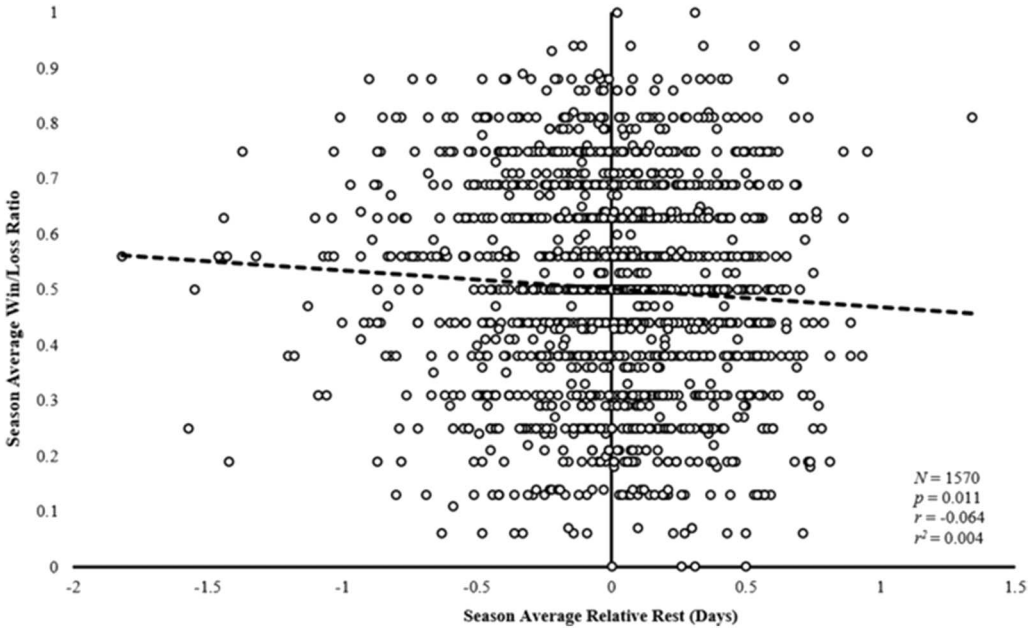

Figure 1. Correlations between season average relative rest index (RRI) and Win/Loss Ratio for all teams.

based on specific RRI ranges (<−0.70; −0.7 to −0.5; −0.5 to −0.25; −0.25 to 0; 0 to 0.25; 0.25 to 0.5; 0.5 to 0.7; >0.70). Total number of teams per RRI category fluctuated because of variable scheduling configuration across seasons and middle categories displaying more common RRI category values. Relative rest index categories and frequency of occurrence of RRI values in each category for all seasons collated and for each decade of NFL competition individually are presented in Table 1.

Statistical Analyses

All statistical analyses were performed in SPSS Version 29.0 (IBM Corporation, Armonk, NY). Pearson Product Correlations between RRI and Win/Loss Ratio were generated for all teams and seasons combined with no specific classification. After assessing all data for normality using a Shapiro–Wilk test, 2 separate 1-way analyses of variance (ANOVAs) with least-significant difference (LSD) pairwise comparisons were also implemented to evaluate the effect of playoff depth (categories 1–6) on RRI and RRI category on Win/Loss Ratio (categories 1–8), respectively. Alpha (p) values were generated for each ANOVA and all pairwise comparisons with statistical significance accepted at $p \leq 0.05$. Effect size for ANOVAs was reported as η_p^2.

Results

All dependent variables met the assumption of normality as evaluated with a Shapiro–Wilk test. Pearson product correlations

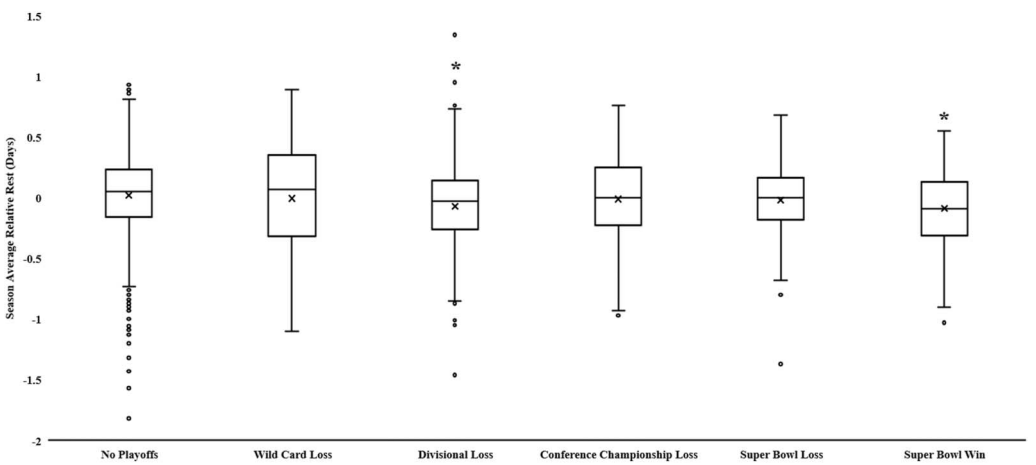

Figure 2. Distribution in relative rest index (RRI) among teams losing in the wild card round, divisional round, conference championship, or Super Bowl, winning the Super Bowl, or not making the playoffs altogether across all seasons. *$p \leq 0.05$ significantly different than no playoffs.

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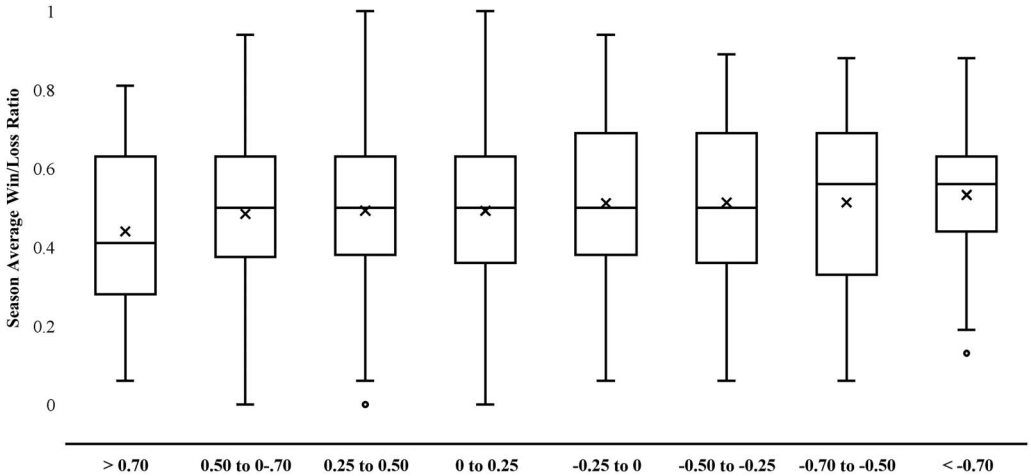

Figure 3. Distribution in Win/Loss Ratio presented across all 8 relative rest index (RRI) categories for all teams and seasons.

between RRI and Win/Loss Ratio for all teams revealed a significant, trivial, negative association ($r = -0.064$, $p = 0.011$, $r^2 = 0.004$) between RRI and Win/Loss Ratio. Correlations between RRI and Win/Loss Ratio for all teams combined are shown in Figure 1.

When evaluating the effect of playoff depth category on RRI, a significant main effect of group ($F = 2.738$, $p = 0.018$, $\eta_p^2 = 0.009$) was observed. Specifically, teams that did not make the playoffs had significantly greater RRI than those that lost in the divisional round ($p = 0.001$) and those that won the Super Bowl ($p = 0.040$). Distribution and effects of RRI values across playoff depth categories are shown in Figure 2. Furthermore, no main effect of group ($F = 1.327$, $p = 0.233$, $\eta_p^2 = 0.006$) was noted for the effect of RRI category on Win/Loss Ratio (Figure 3).

Discussion

Based on our data, RRI and Win/Loss Ratio were not strongly correlated across 52 seasons of NFL competition. Contrary to our hypothesis, RRI was significantly influenced by playoff performance such that teams advancing to the divisional round or winning the Super Bowl had impaired RRI compared with those not qualifying for the playoffs. Interestingly, Win/Loss Ratio remained unchanged across varying RRI categories. Collectively, these data indicate that increased RRI does not significantly enhance a team's chance of winning a given NFL competition.

One previous investigation from our laboratory has evaluated the effect of RRI on Win/Loss Ratio in the NFL (12). This study evaluated this relationship during the 2022–23 NFL season and reported significant positive correlations between RRI and Win/Loss Ratio such that changes in RRI could account for up to 11.4% of the variability in each team's Win/Loss Ratio, differing greatly from our findings (12). Results from the present investigation were surprising given an abundance of evidence that athletic competition impairs performance and recovery by altering inflammation and muscle damage acutely and throughout an entire season across multiple sports (3,5,11,19). Specifically, an American Football game has been demonstrated to alter athletic performance and inflammation, while attenuating perceived recovery acutely and up to 4 days postcompetition, lending support to the idea that demand accumulates during the course of a competitive football season (6,8,9). Most unexpectedly, however, our data indicate that teams that progress further in playoff competition have impaired RRI compared with those not qualifying for the playoffs altogether. Based on this finding, teams that progress further in playoff competition seem to overcome any potential disadvantage imposed by decreased RRI. With the advent of Thursday Night Football in 2006 and increased frequency of prime-time games in the 1990's and early 2000's, RRI has varied more heavily in recent years as given in Table 1 (4,14). This concomitant variability in scheduling and lack of longitudinal relationship between RRI and Win/Loss Ratio supports the notion that RRI may not be a major predictor of performance among elite NFL teams.

Based on these data, scheduling configuration in the NFL does not seem to have robust effects on team performance. However, this analysis is not without limitations. Multiple confounding factors can modulate the relationship between RRI and team performance including team skill level, injuries at specific times of the season, and individual variability in recovery strategies and practice and training schedules between teams. Unfortunately, much of these data were not publicly available, making their inclusion in this study challenging. Although this study expanded on the findings presented by Jajtner et al. (12) by including more competitive seasons in our analysis, a more sensitive analysis is warranted. As teams progressing further in the playoffs had impaired RRI while maintaining favorable Win/Loss Ratios, it is plausible that RRI fluctuations are more impactful among lesser teams. This study did not evaluate the effect of RRI on team performance completely independent of teams that progress furthest in playoff competition, potentially raising concerns regarding the sensitivity of our analysis to reveal these specific effects. In addition, no analysis specifically compared team performance among varying RRI categories across eras of NFL competition despite our data clearly demonstrating greater occurrence of highly variable season average RRIs in recent years. As such, further work should investigate variability in team

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performance across RRI categories and across all seasons of NFL competition with eras compared.

Practical Applications

These data demonstrate that RRI does not significantly affect Win/Loss Ratio across all seasons of NFL competition from 1970 to the present day. In fact, teams progressing further in competition displayed reduced RRI. As such, the present scheduling configuration used by the NFL seems to maintain competitive equity. However, this analysis focused solely on team performance. Coaches and practitioners should not disregard or underprioritize player recovery from both practice and competition because of the robust demands imposed by an American Football season.

Acknowledgments

Conception and design of research: A. G. Pinzone, A. R. Jajtner, J. E. Barkley; acquisition of data: A. G. Pinzone, A. R. Jajtner, Y. Abu-Amara; data analysis and interpretation: A. G. Pinzone, A. R. Jajtner, R. W. Gant, J. E. Barkley; manuscript draft and revisions: A. G. Pinzone, A. R. Jajtner, Y. Abu-Amara, R. W. Gant, J. Rivera, E. C. Tagesen, J. E. Barkley; approval of final version: A. G. Pinzone, A. R. Jajtner, Y. Abu-Amara, R. W. Gant, J. Rivera, E. C. Tagesen, J. E. Barkley. The authors have no conflicts of interest to disclose.

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